2026aj05355 ClassifyLab — Submission Guide & README

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Final submission items

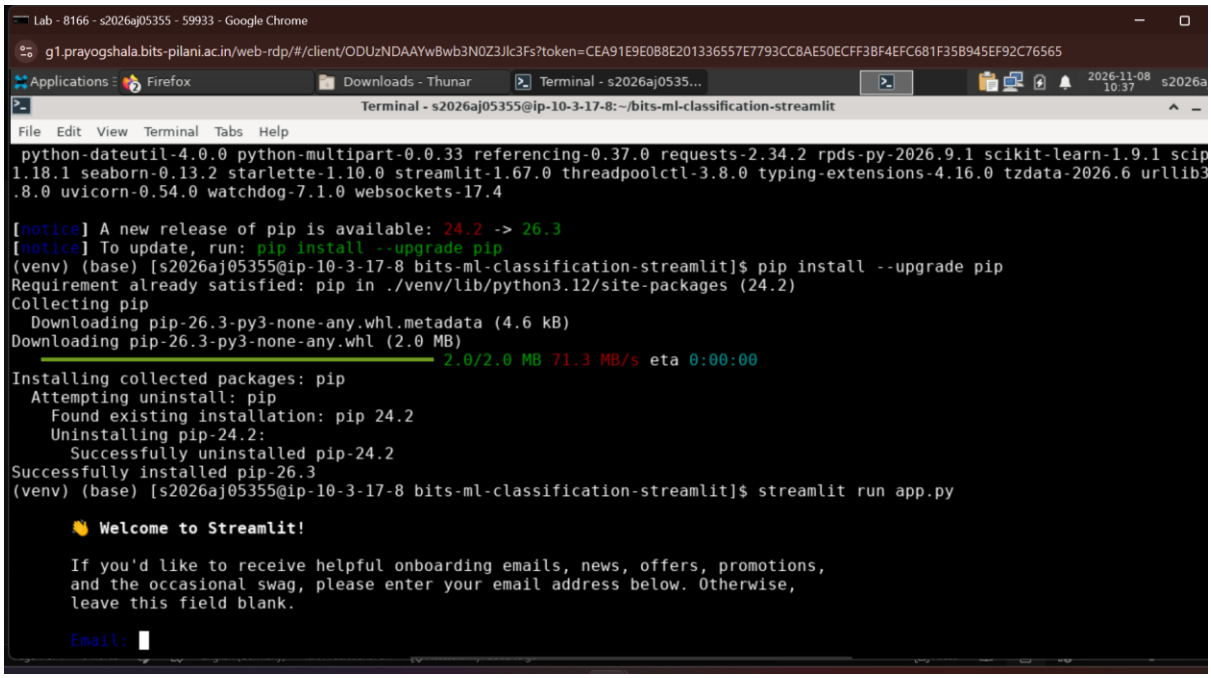
1. GitHub repository link: [jeenaraju-sys/bits-ml-classification-streamlit: Breast Cancer Classification — Streamlit ML Assignment]

2. Live Streamlit Community Cloud link: [ClassifyLab | ML Model Studio · Streamlit]

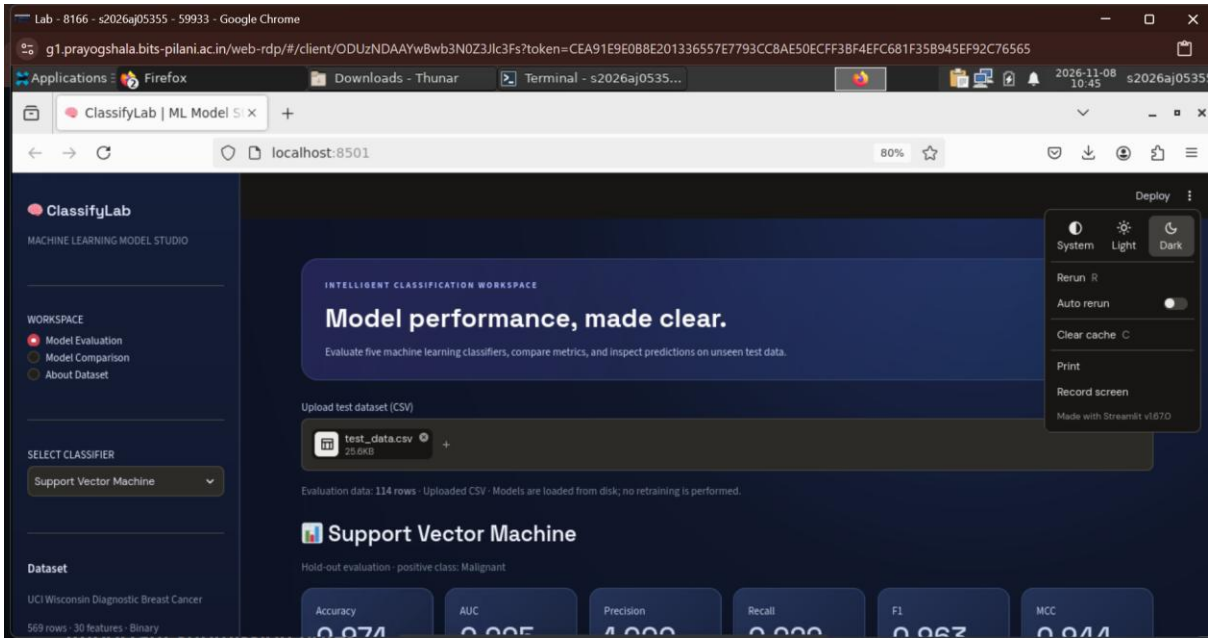
3. Insert the own screenshot of the assignment running in BITS Virtual Lab.

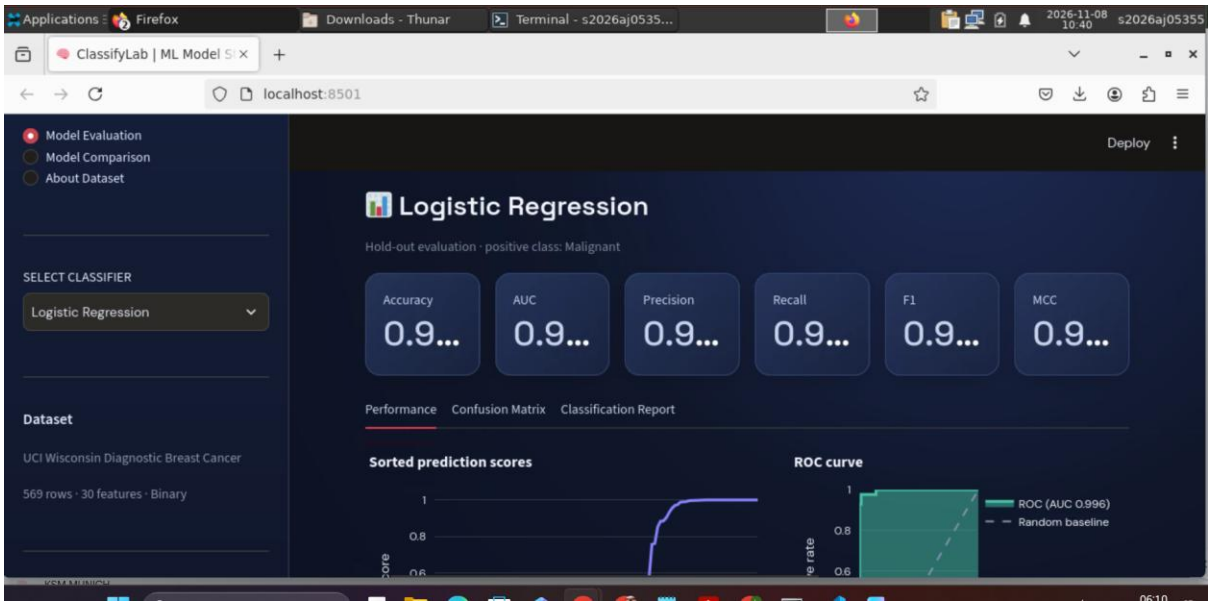
4. README content follows in this PDF.

BITS Virtual Lab screenshot

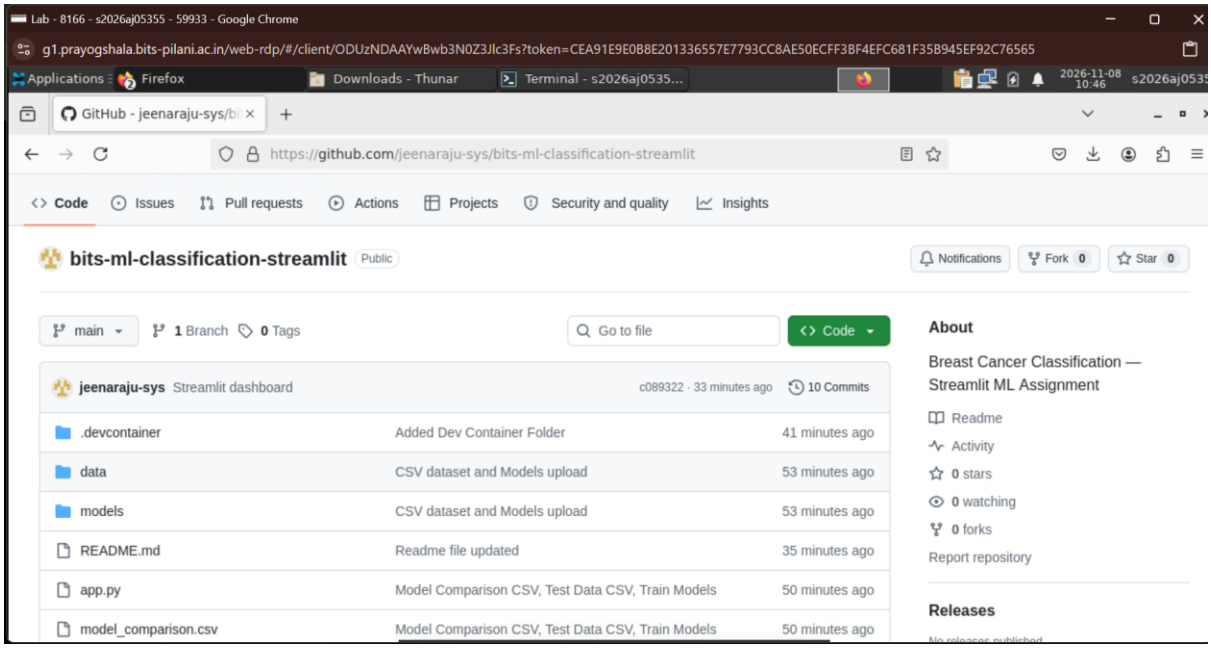


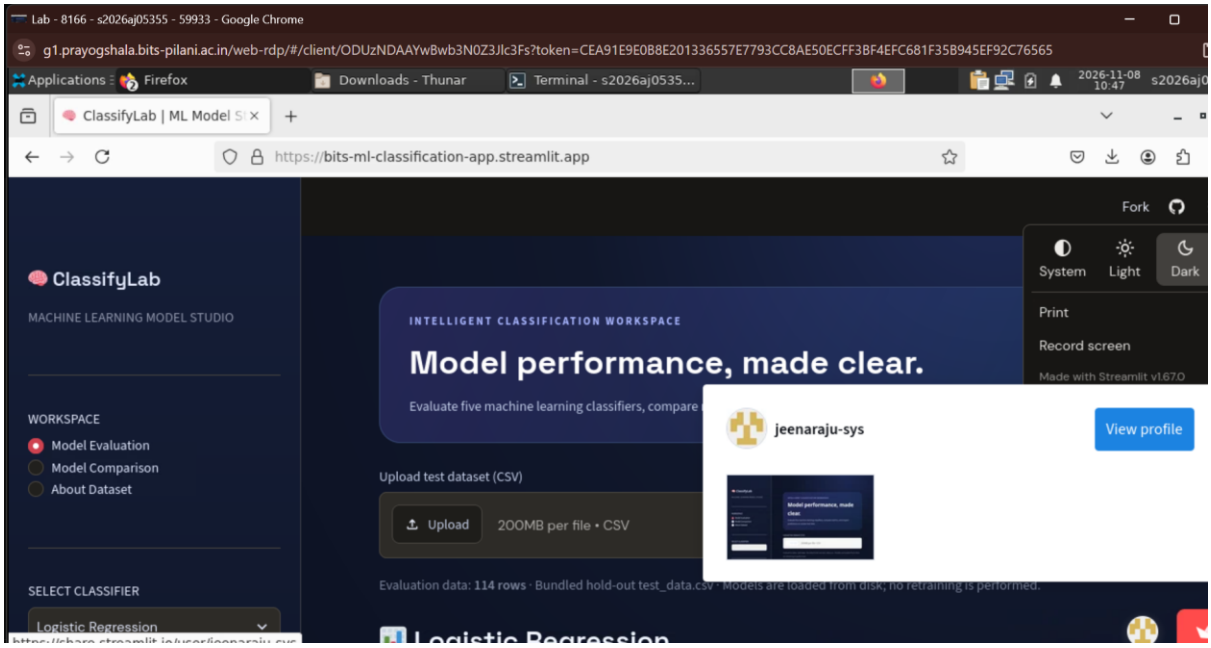
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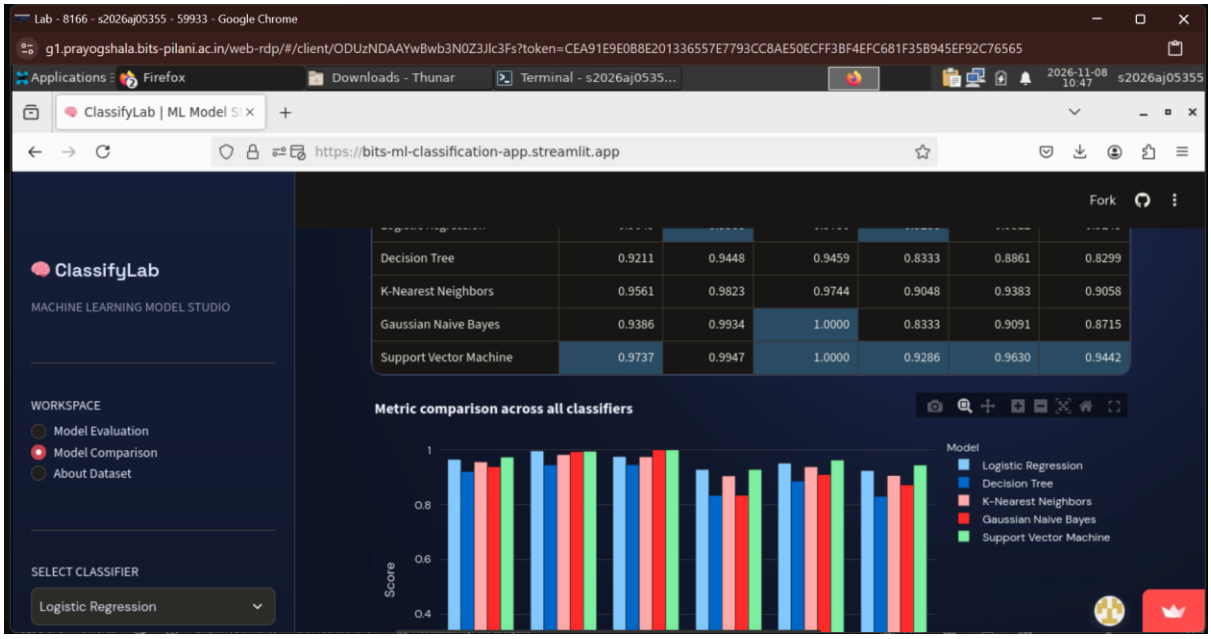


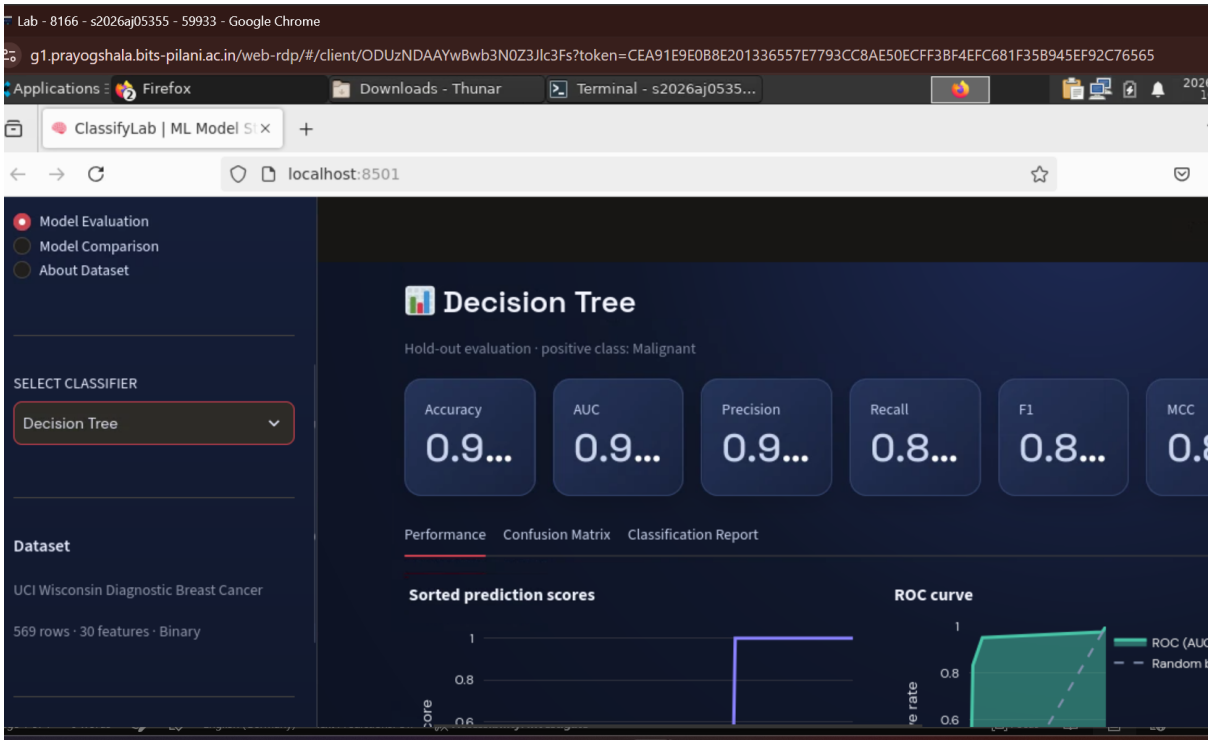
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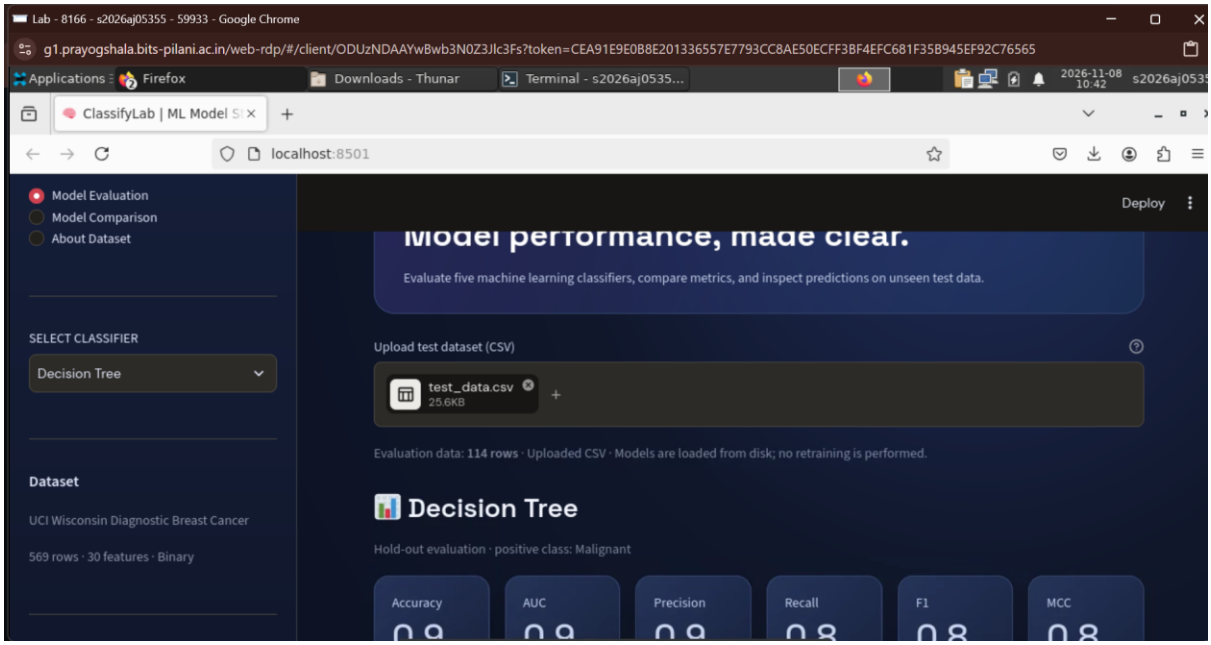


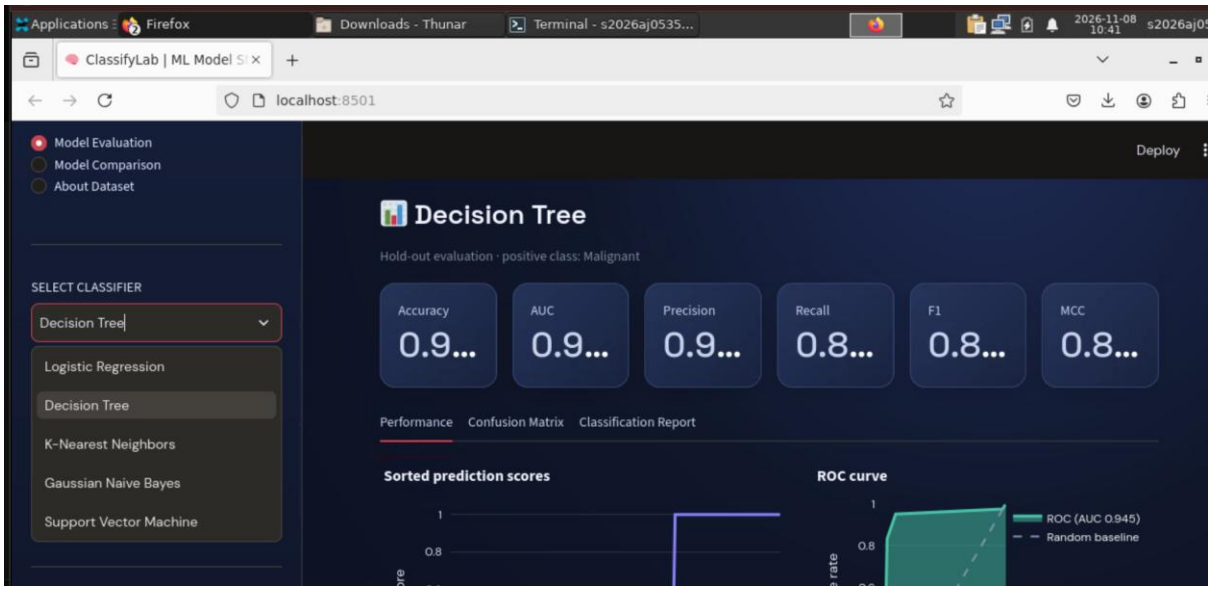
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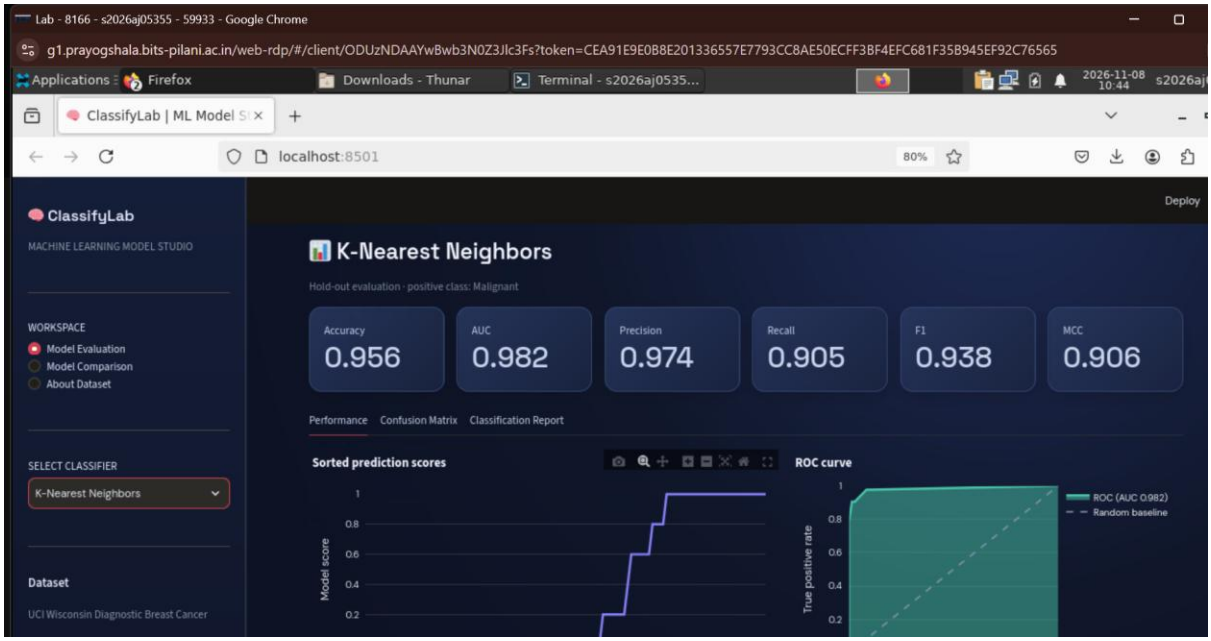


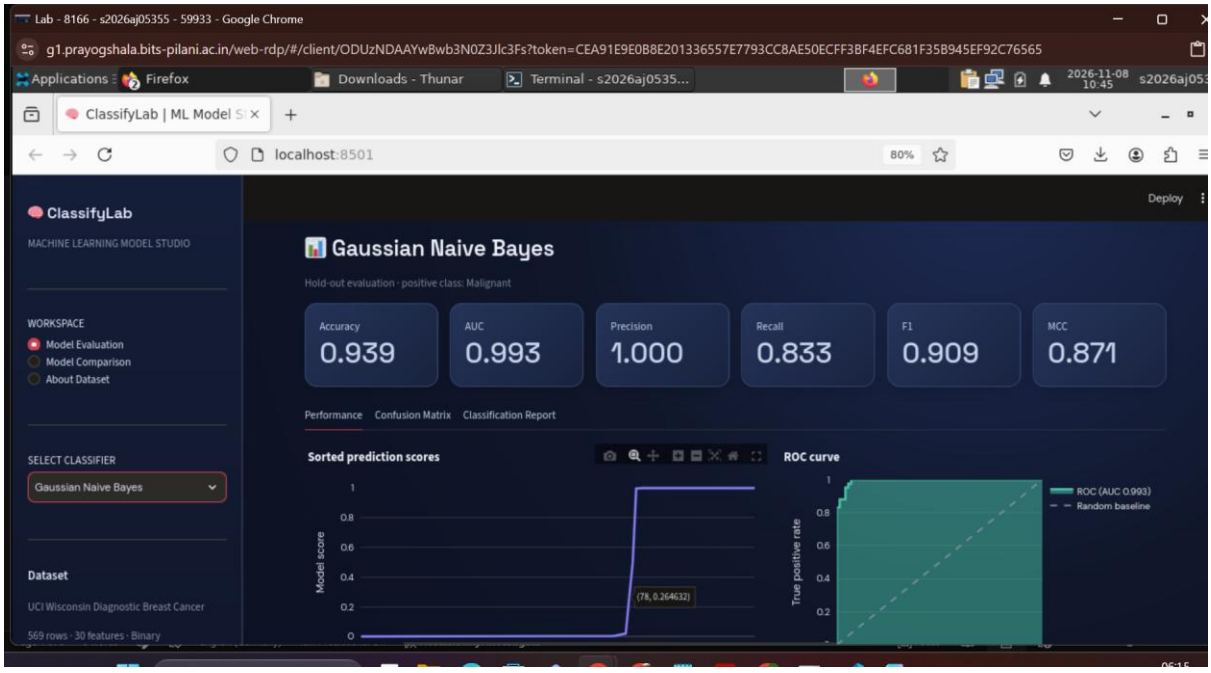
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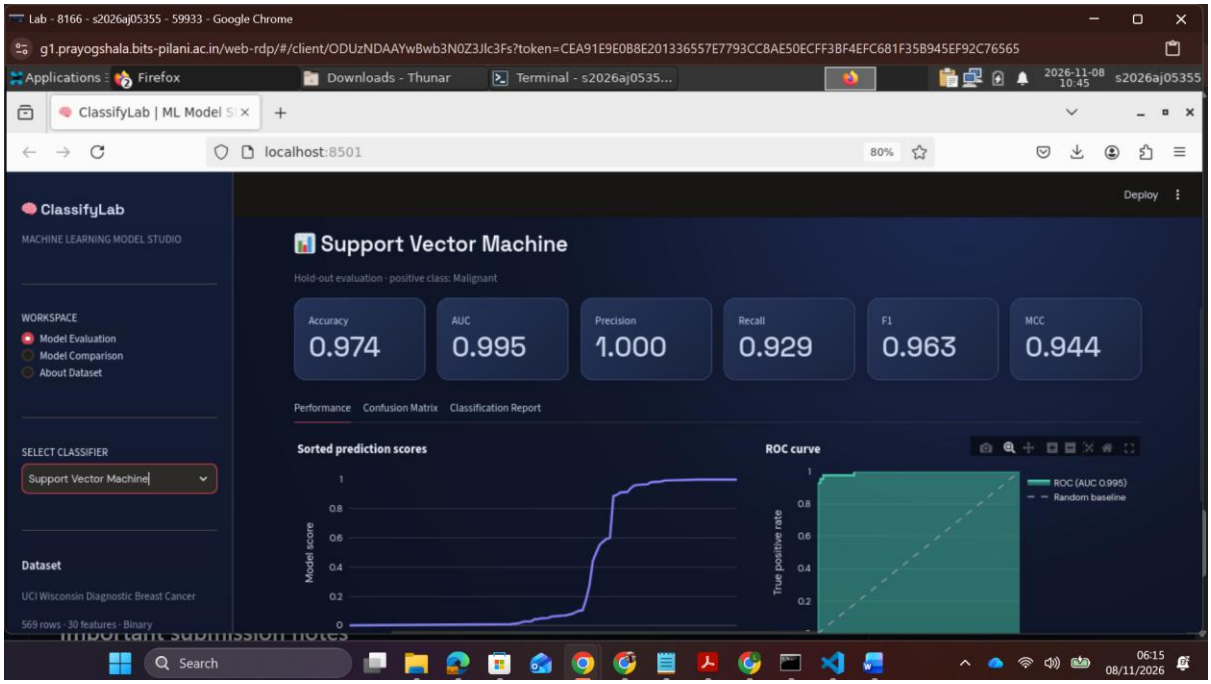


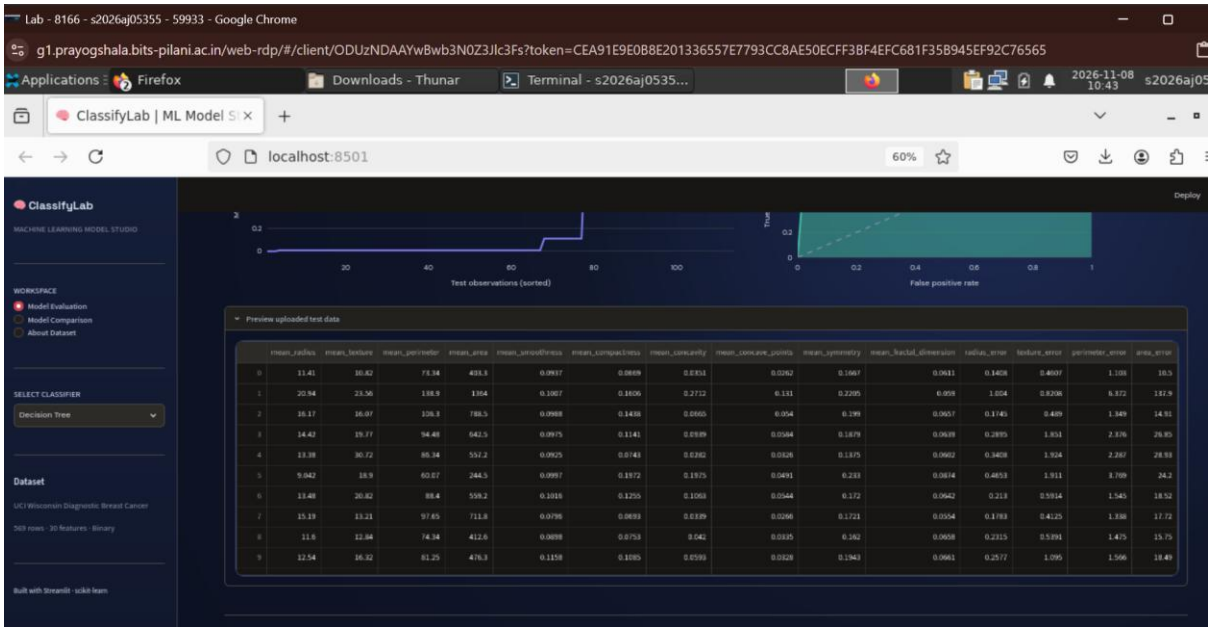
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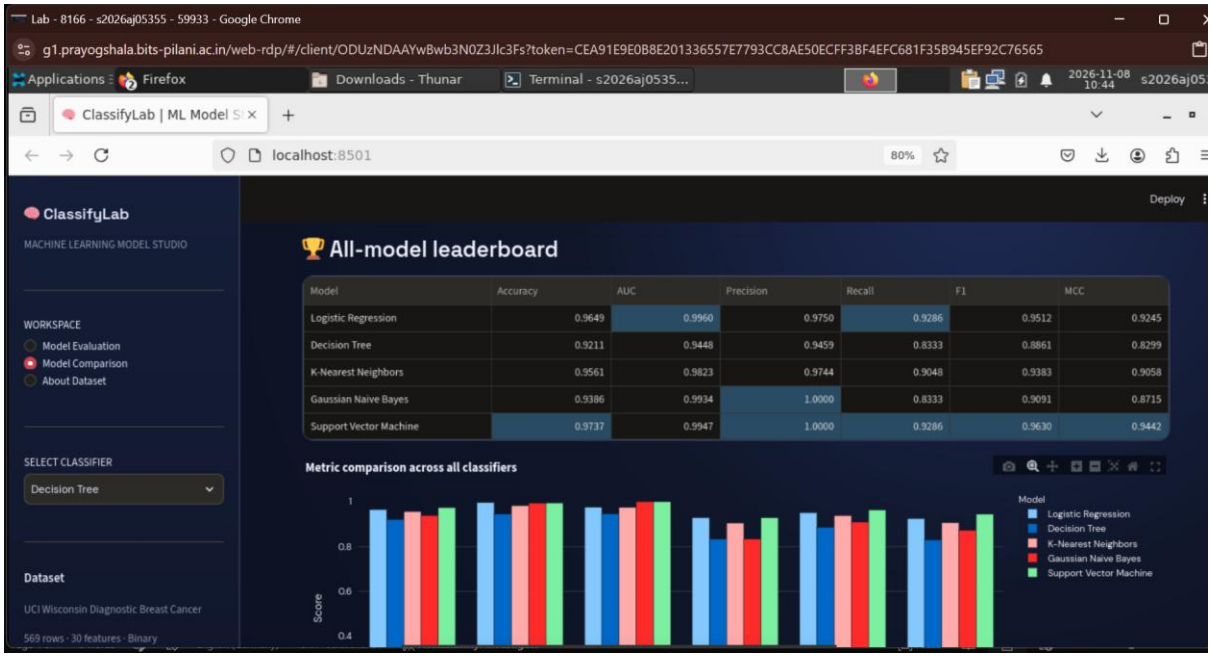


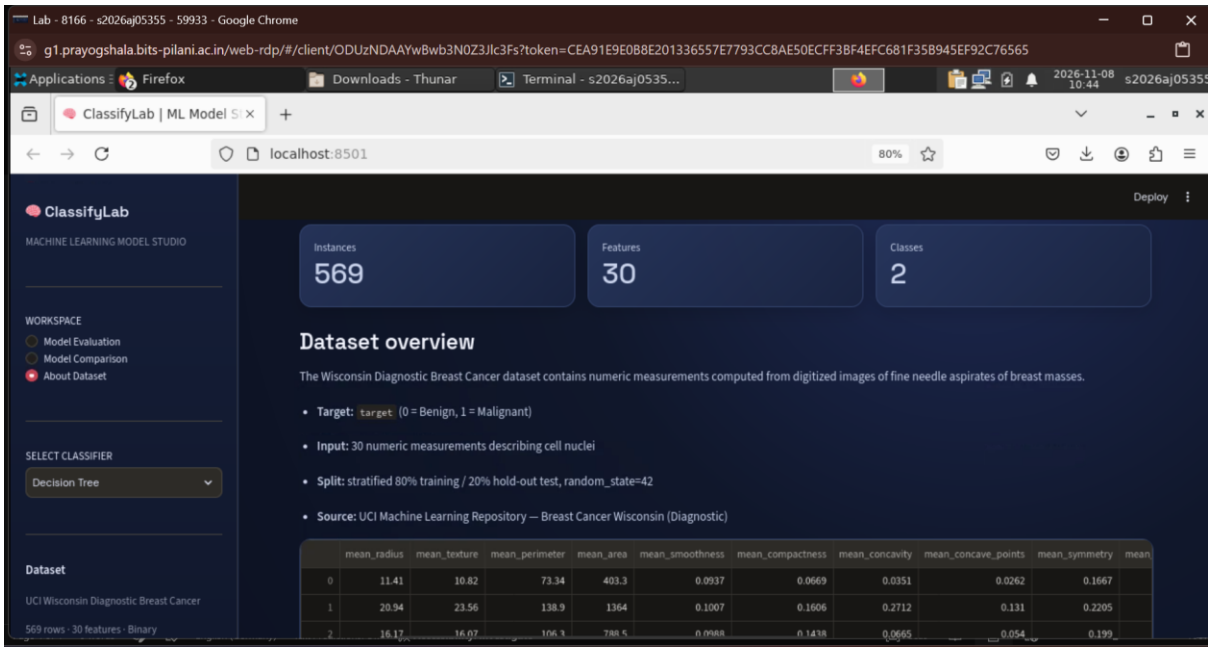
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a. Problem statement

Build and deploy an interactive machine-learning classification application that compares five classifiers on one common dataset. The applicationloads pre-trained models, accepts a labeled test CSV, computes six evaluation metrics, and displays a confusion matrix, classification report, ROC curve, and comparison charts. Models are not retrained when a user uploads test data.

b. Dataset description

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- **Dataset:** Breast Cancer Wisconsin (Diagnostic), UCI Machine Learning Repository.
- **Public source:** https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic
- **Instances:** 569.
- **Input features:** 30 numeric features.
- **Task:** Binary classification.
- **Target encoding**: target = 0 means Benign; target = 1 means Malignant.
- **Split:** Stratified 80% training / 20% hold-out test, random_state=42.
- The project includes data/wdbc_dataset.csv and the held-out test_data.csv.

c. GitHub Repository Link

Add the public GitHub repository URL here after pushing this project:

GitHub: [jeenaraju-sys/bits-ml-classification-streamlit: Breast Cancer Classification — Streamlit ML Assignment]

d. Models used and evaluation

All five models are trained on the same training partition and evaluated on the same held-out test partition. The app recalculates these metrics for

the selected uploaded CSV.

Comparison table

Comparison table

Model	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9649	0.9960	0.9750	0.9286	0.9512	0.9245
Decision Tree	0.9211	0.9448	0.9459	0.8333	0.8861	0.8299
K-Nearest Neighbors	0.9561	0.9823	0.9744	0.9048	0.9383	0.9058
Gaussian Naive Bayes	0.9386	0.9934	1.0000	0.8333	0.9091	0.8715
Support Vector Machine	0.9737	0.9947	1.0000	0.9286	0.9630	0.9442

Observations

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ML Model Name	Observation about model performance
Logistic Regression	Excellent balanced performance (0.965 accuracy, 0.996 AUC, 0.951 F1). Precision is 0.975 and recall is 0.929, so it makes few false-positive predictions while detecting most malignant cases.
Decision Tree	Lowest overall scores on this split (0.921 accuracy, 0.886 F1, 0.830 MCC). It is easier to interpret, but the depth-limited tree misses more malignant cases (0.833 recall).
K-Nearest Neighbors	Strong performance (0.956 accuracy, 0.982 AUC, 0.938 F1). It offers a good balance, although its scores are slightly below Logistic Regression and SVM on this split.
Gaussian Naive Bayes	Achieves perfect precision (1.000) but lower recall (0.833), meaning its malignant predictions are reliable but it misses more actual malignant cases than the leading models.
Support Vector Machine	Best overall on this split: 0.974 accuracy, 0.963 F1, and 0.944 MCC. Precision is 1.000 and recall is 0.929, giving the strongest combined result among the five models.

Overall winner for this dataset: Support Vector Machine on the supplied 20% hold-out test split, selected by F1 first, then AUC and MCC as tie-breakers. This is a split-specific result, not a guarantee of future performance.

Metrics

• **Accuracy**: fraction of all predictions that are correct.

• **AUC:** area under the ROC curve, using model probability/decision scores.

• **Precision**: fraction of predicted malignant cases that are actually malignant.

•**Recall:** fraction of actual malignant cases correctly identified.

• **F1:** harmonic mean of precision and recall.

• **MCC:** correlation coefficient between observed and predicted classes.

Application features

• CSV upload for labeled test data (30 required feature columns plus target).

• Left sidebar model selector and workspace navigation.

• Single-model evaluation and all-model comparison.

• Six metric cards, comparison table, interactive Plotly charts, ROC curve, confusion matrix, and classification report.

• Five saved .joblib model files. No training occurs in the deployed app.

Run locally

1. Create and activate a virtual environment (recommended)

python -m venv .venv

Windows:

.venv\Scripts\activate

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```
# macOS/Linux:
source .venv/bin/activate
# 2. Install dependencies
pip install -r requirements.txt
# 3. Start the dashboard
streamlit run app.py
```

Open the local URL printed by Streamlit, usually http://localhost:8501.

To recreate the saved models:

python train_models.py

Deploy on Streamlit Community Cloud (FREE)

1. Create/sign in to GitHub and create a **public** repository.

2. Upload all project files and folders from this project. Ensure app.py, requirements.txt, README.md, test_data.csv, and models/ are at the

repository root.

3. Visit https://share.streamlit.io/ (or https://streamlit.io/cloud) and sign in with GitHub.

4. Choose **Create app / New app**, select the repository and main branch.

5. Set the main file path to app.py, then deploy.

6. Wait for the build to finish. Open the app URL and test the sidebar, comparison page, model metrics, charts, and CSV upload.

7. **Live Streamlit App:** ClassifyLab | ML Model Studio · Streamlit

Suggested repository structure

```
classification_dashboard_project/
■■■ app.py
■■■ train_models.py
■■■ requirements.txt
■■■ README.md
■■■ test_data.csv
■■■ model_comparison.csv
■■■ models/
■   ■■■ logistic_regression.joblib
■   ■■■ decision_tree.joblib
■   ■■■ knn.joblib
■   ■■■ naive_bayes.joblib
■   ■■■ svm.joblib
■■■ data/
    ■■■ wdbc_dataset.csv
```

Limitations and responsible use

This is an educational classification exercise using a public medical dataset. It is not a medical device and must not be used to diagnose, screen,

or make clinical decisions.